

Lecture

SARIMA(p, d, q) x (P, D, Q)

$$X_t = \alpha_0 + \alpha_1 X_{t-1} + \dots + \alpha_p X_{t-p} + Z_t + \beta_1 Z_{t-1} + \dots + \beta_q Z_{t-q} + \\ + \alpha_S X_{t-S} + \alpha_{2S} X_{t-2S} + \dots + \alpha_{PS} X_{t-PS} + \\ + \beta_S Z_{t-S} + \beta_{2S} Z_{t-2S} + \dots + \beta_{QS} Z_{t-QS}$$

where $\alpha_0 = \mu(1 - \alpha_1 - \dots - \alpha_p)$

Reminder about sample ACVF and ACF

Estimating parameters

- α_0, α, β
- d, D
- q, Q
- p, P

Estimating the mean

$$\hat{\mu} = \frac{1}{T} \sum_{t=1}^T X_t$$

Must be taken with care:

- Only makes sense for stationary processes
- If there is autocorrelation, the variance of the estimator depends on it
- Not necessarily that $\hat{\mu} \rightarrow \mathbb{E}[X_t]$, requires property called **ergodicity**, usually OK for stationary processes

Estimating AR process

Consider $AR(p)$ process

$$X_t = \alpha_0 + \alpha_1 X_{t-1} + \dots + \alpha_p X_{t-p} + Z_t$$

Note that it is linear in X_t . Thanks to that, we have many ways to do the estimation:

- Least squares:

$$\min_{\alpha_0, \alpha_1, \dots, \alpha_p} \sum_{t=p+1}^T [X_t - \alpha_0 - \alpha_1 X_{t-1} - \dots - \alpha_p X_{t-p}]^2$$

- Maximum likelihood:

$$\min_{\alpha_0, \alpha_1, \dots, \alpha_p} \log \mathcal{L}(\alpha|X)$$

$$\mathcal{L}(\alpha|X) = \left(\prod_{t=p+1}^T p(X_t | X_{t-1}, \dots, X_{p+1}, \theta) \right) p(X_1, \dots, X_p | \theta)$$

$$\log \mathcal{L}(\alpha|X) = \underbrace{\sum_{t=p+1}^T \log p(X_t | X_{t-1}, \dots, X_{p+1}, \theta)}_{\text{conditional ll}} + \underbrace{\log p(X_1, \dots, X_p | \theta)}_{\text{marginal ll}}$$

Will coincide with LSE if the noise is from Normal distribution.

Estimating AR process

Only good for large N and strong stationary properties:

- As ordinary regression treating each X_{t-k} as independent variable:

$$X_t \sim \alpha_0 + \alpha_1 X_{t-1} + \dots + \alpha_p X_{t-p} + Z_t$$

- Using Yule-Walker equations:

$$\rho(k) = \alpha_1 \rho(k-1) + \dots + \alpha_p \rho(k-p)$$

$$R\hat{\alpha} = r$$

where r is a vector of sample autocorrelations and R is a matrix of them.

Example: AR(1)

Consider $AR(1)$ process

$$X_t = \mu + \alpha_1 X_{t-1} + Z_t$$

The estimation using least squares will give:

$$\hat{\mu} = \frac{\bar{X}_{(2)} - \hat{\alpha}_1 \bar{X}_{(1)}}{1 - \hat{\alpha}_1}$$

where $\bar{X}_{(1)}$ and $\bar{X}_{(2)}$ are the means of the first and last $(N - 1)$ observations respectively. Obviously they are close, so often used estimator is the sample mean:

$$\hat{\mu} \approx \bar{X}$$

$$\hat{\alpha}_1 = \frac{\sum_{t=1}^{N-1} (X_t - \hat{\mu})(X_{t+1} - \hat{\mu})}{\sum_{t=1}^{N-1} (X_t - \hat{\mu})^2} \approx \frac{\sum_{t=1}^{N-1} (X_t - \bar{X})(X_{t+1} - \bar{X})}{\sum_{t=1}^{N-1} (X_t - \bar{X})^2} \approx \frac{\sum_{t=1}^{N-1} (X_t - \bar{X})(X_{t+1} - \bar{X})}{\sum_{t=1}^N (X_t - \bar{X})^2}$$

Estimating MA process

Consider $MA(q)$ process:

$$X_t = \alpha_0 + \beta_1 Z_{t-1} + \dots + \beta_q Z_{t-q}$$

Unfortunately, for such a process, we can not express the residual as a linear function of observed X_t s. Hence, only the maximum likelihood method is applicable.

$$\min_{\alpha_0, \alpha_1, \dots, \alpha_p} \log \mathcal{L}(\alpha|X)$$

$$\mathcal{L}(\alpha|X) = \left(\prod_{t=p+1}^T p(X_t|X_{t-1}, \dots, X_{p+1}, \theta) \right) p(X_1, \dots, X_p|\theta)$$

$$\log \mathcal{L}(\alpha|X) = \underbrace{\sum_{t=p+1}^T \log p(X_t|X_{t-1}, \dots, X_{p+1}, \theta)}_{\text{conditional ll}} + \underbrace{\log p(X_1, \dots, X_p|\theta)}_{\text{marginal ll}}$$

Estimating ARMA process

As a composition of $AR(p)$ and $MA(q)$, $ARMA(p, q)$ should be estimated with maximum likelihood method. Optimization of conditional likelihood is what happens under the hood of most practical methods.

Estimating differencing order

- d and D are chosen such that the series is stationary.
- If there is seasonality, it is recommended to start with seasonal differencing, it may already make the series stationary, as we've seen last time.
- The less differencing, the better, because
 - More data

- Less variance due to back transformation

Estimating MA and AR order

- We absolutely can't estimate p and q with maximum likelihood, the higher the order, the higher $\mathcal{L}$
- We can select the order by looking at ACF and PACF
- We can select the order automatically with information criteria

PACF

Partial autocorrelation function is defined as follows:

$$\pi(h, h) = \begin{cases} \rho(X_{t+1}, X_t), & h = 1, \\ \rho(X_{t+h} - X_{t+h}^{h-1}, X_t - X_t^{h-1}), & h > 1 \end{cases}$$

where X_t^{h-1} is a result of regressing X_t on its $h - 1$ previous values.

In words: When fitting an $AR(p)$ model, the last coefficient α_p will be denoted by π_p and measures the excess correlation at lag p which is not accounted for by an $AR(p - 1)$ model. It is called the p -th partial autocorrelation coefficient and, when plotted against p , gives the partial ACF.

Estimating MA and AR order with ACF and PACF

- We know that for $MA(q)$ process, the ACF becomes zero after q lags.
- We also know that for $AR(p)$ process, the ACF is a mixture of exponential decay and sinusoid.
- It can be shown that the PACF has reverse properties to those above.

So by looking at ACF we define the order q and by looking at PACF we define the order p .

Also the seasonal order can be defined, the lags will be significant at $P \cdot S$ and $Q \cdot S$.

Estimating MA and AR order with information criteria

Information criteria:

- Akaike

$$AIC = -2\mathcal{L} + 2k$$

- Corrected Akaike (for small samples)

$$AICc = -2\mathcal{L} + \frac{2k^2}{T - k - 1}$$

- Bayesian

$$BIC = -2\mathcal{L} + k(\log T - 2)$$

here $k = P + Q + p + q + 1$ is the number of parameters in the model.

Residuals

When a model has been fitted to a time series, it is advisable to check that the model really does provide an adequate description of the data. As with most statistical models, this is usually done by looking at the residuals:

$$\text{residual} = \text{observation} - \text{fitted value}$$

For a univariate time-series model, the fitted value is the one-step-ahead forecast so that the residual is the one-step-ahead forecast error.

The properties of residuals, e.g. lack of structure, quantify the goodness of the model.

Necessary properties of residuals

- Unbiasedness; to be checked with standard statistical tests for the mean, e.g. Student's t-test
- Stationarity; to be checked with visual analysis, KPSS test
- Uncorrelatedness; to be checked with correlogram, Ljung-Box-Pierce Q-criterion

Desired properties of residuals

- Normality; to be checked with QQ-plot, Shapiro test
- Homoscedacity; to be checked with visual analysis, Breusch-Pagan test

Webinar

```
In [1]: import numpy as np
```

```
import IPython.display as dp
import matplotlib.pyplot as plt
import seaborn as sns

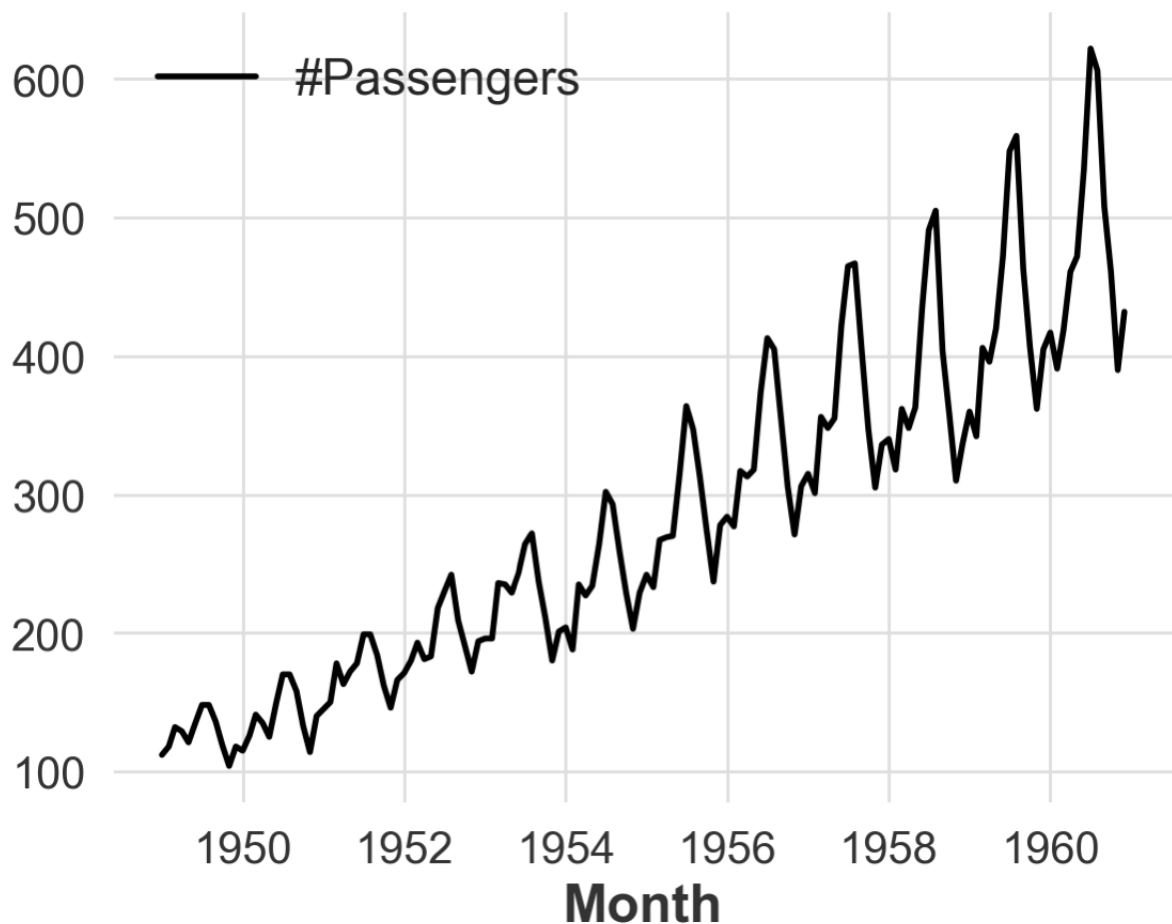
dp.set_matplotlib_formats("retina")
sns.set(style="whitegrid", font_scale=1.5)
sns.despine()

%matplotlib inline
```

```
/var/folders/33/j0cl7y453td68qb96j7bqcj4cf41kc/T/ipykernel_35952/402538753
1.py:7: DeprecationWarning: `set_matplotlib_formats` is deprecated since I
Python 7.23, directly use `matplotlib_inline.backend_inline.set_matplotlib
_formats()`
  dp.set_matplotlib_formats("retina")
<Figure size 640x480 with 0 Axes>
```

```
In [2]: from darts import TimeSeries
import darts.datasets as ds
import scipy.stats as sts
import statsmodels.api as sm
import statsmodels.tsa.api as tsa
from darts.utils.statistics import plot_acf
```

```
In [3]: air_pax = ds.AirPassengersDataset().load()
air_pax.plot();
```




```
/Users/nstulov/miniconda3/envs/msai/lib/python3.12/site-packages/statsmodels/tsa/statespace/sarimax.py:978: UserWarning: Non-invertible starting MA parameters found. Using zeros as starting parameters.  
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/Users/nstulov/miniconda3/envs/msai/lib/python3.12/site-packages/statsmodels/base/model.py:607: ConvergenceWarning: Maximum Likelihood optimization failed to converge. Check mle_retvals  
warnings.warn("Maximum Likelihood optimization failed to "  
/Users/nstulov/miniconda3/envs/msai/lib/python3.12/site-packages/statsmodels/tsa/statespace/sarimax.py:978: UserWarning: Non-invertible starting MA parameters found. Using zeros as starting parameters.  
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```

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[illegible]

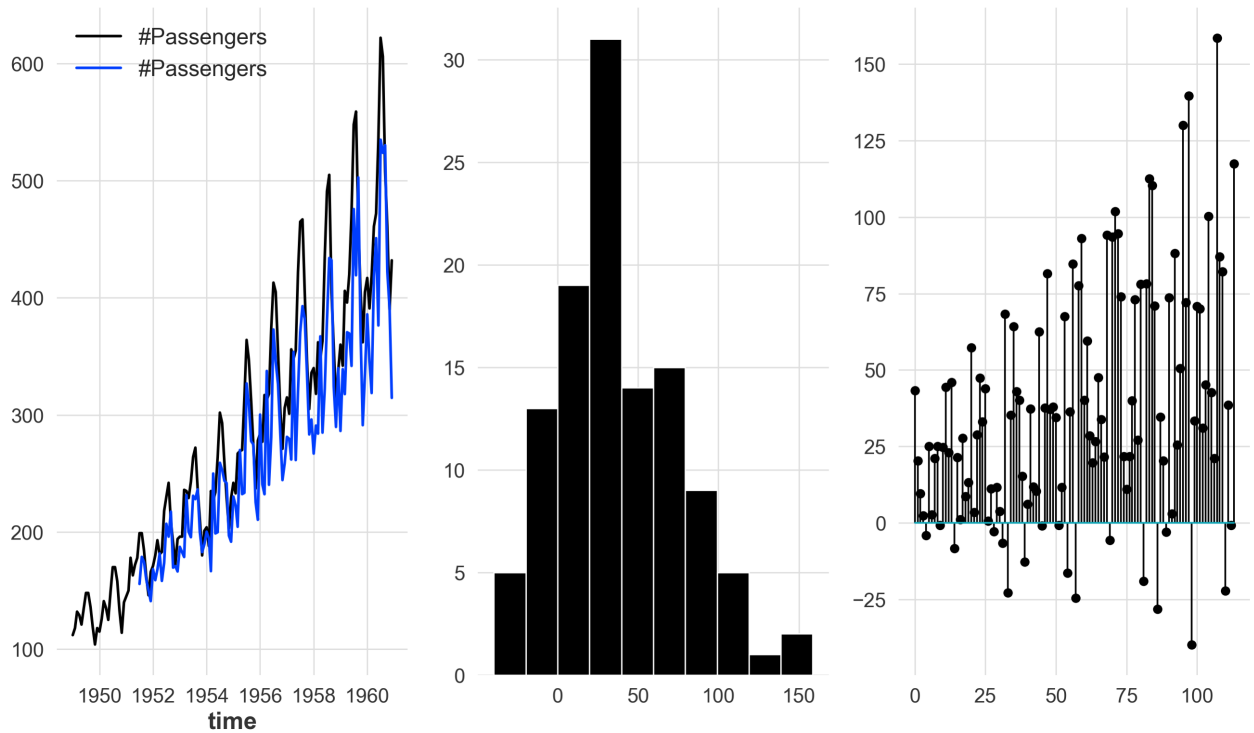
```

parameters found. Using zeros as starting parameters.
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warn('Non-invertible starting MA parameters found.')

```

```
In [10]: resid = air_pax[fcast.start_time():] - fcast
```

```
In [11]: fig, ax = plt.subplots(1,3,figsize=(16,9))
air_pax.plot(ax=ax[0]);
fcast.plot(ax=ax[0]);
ax[1].hist(resid.values());
ax[2].stem(resid.values());
```



```
In [12]: pvalue = sts.ttest_1samp(resid.values(), 0).pvalue
         rejected = pvalue < 0.05
         print("zero mean" if not rejected else "non-zero mean")
```

non-zero mean

Necessary properties of residuals: stationarity

```
In [13]: unst_model = ARIMA(p=0, d=0, q=1, seasonal_order=(0, 0, 0, 0))
         fcast = unst_model.historical_forecasts(
             air_pax, forecast_horizon=1, start_format="position", show_warnings=F
         )
         resid = air_pax[fcast.start_time():] - fcast
```

```
/Users/nstulov/miniconda3/envs/msai/lib/python3.12/site-packages/statsmodels
tsa/statespace/sarimax.py:978: UserWarning: Non-invertible starting MA
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/Users/nstulov/miniconda3/envs/msai/lib/python3.12/site-packages/statsmodels
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```

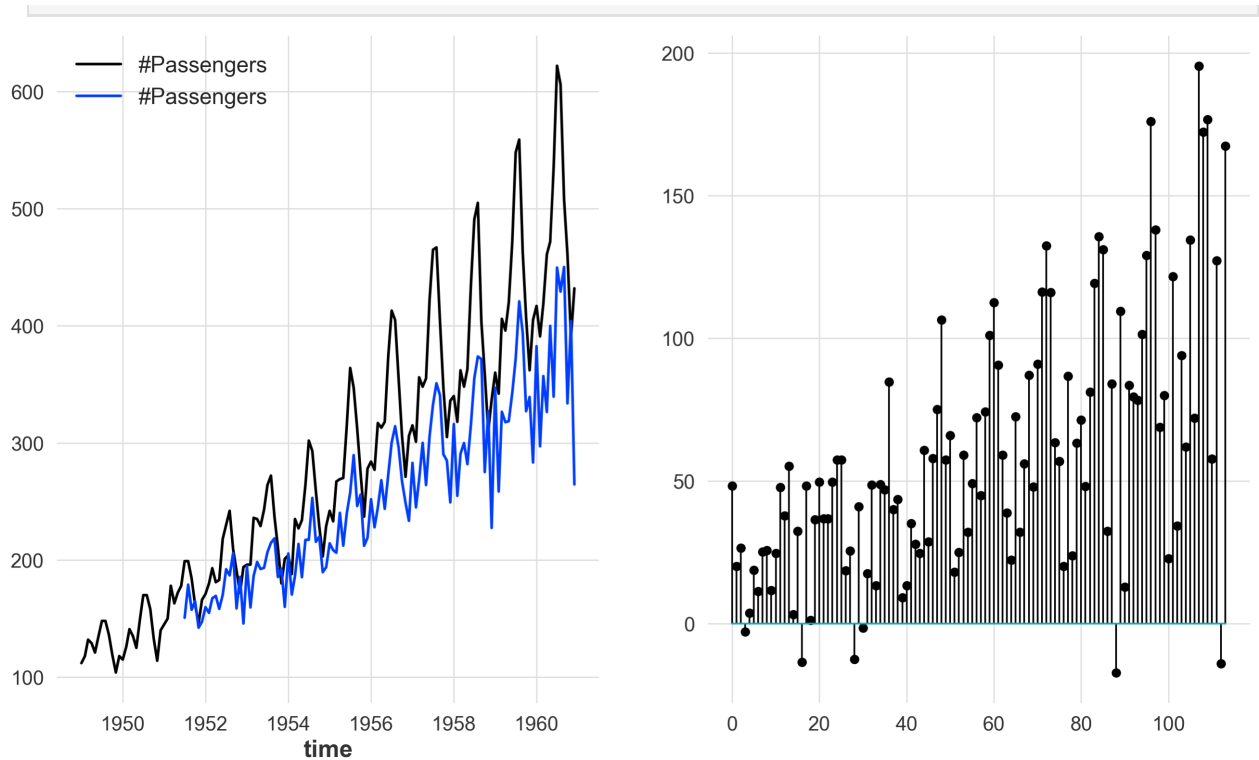
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warn('Non-invertible starting MA parameters found.')

```

```

In [14]: fig, ax = plt.subplots(1,2,figsize=(16,9))
air_pax.plot(ax=ax[0]);
fcast.plot(ax=ax[0]);
ax[1].stem(resid.values());

```



```
In [15]: pvalue = tsa.stattools.kpss(resid.values())[1]
         rejected = pvalue < 0.05
         print("stationary" if not rejected else "non-stationary")
```

non-stationary

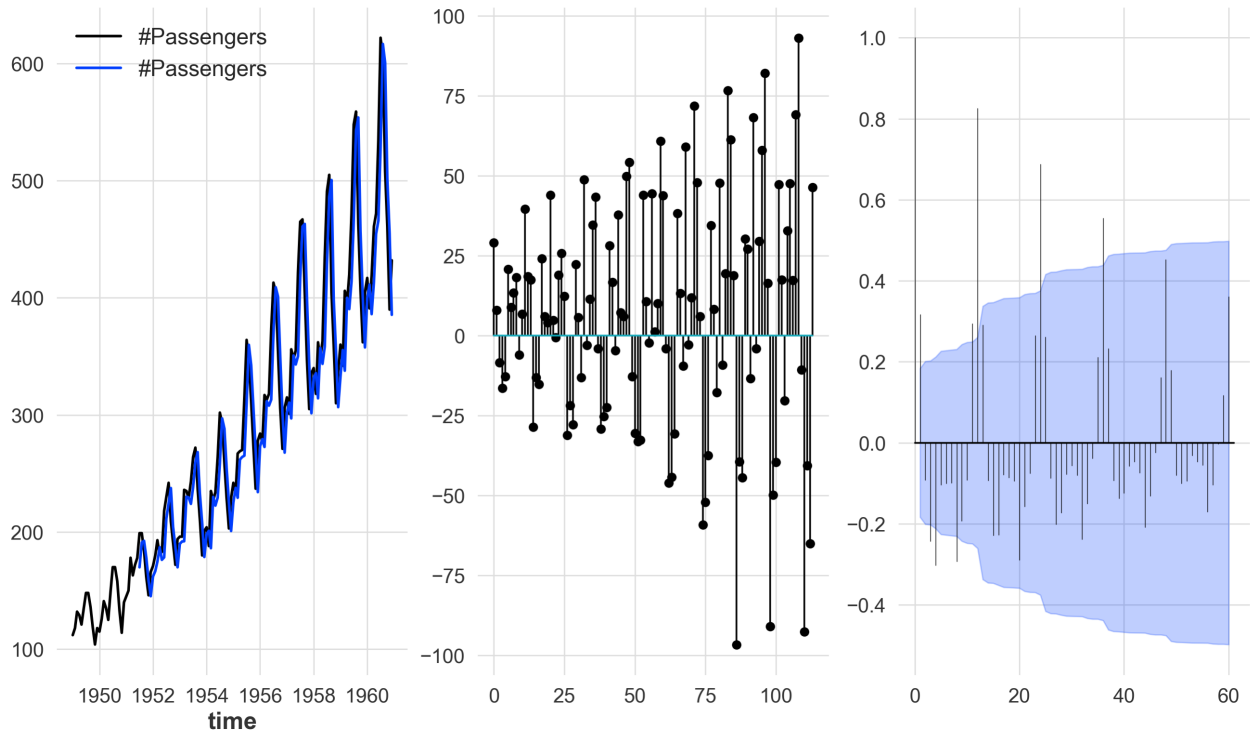
/var/folders/33/j0cl7y453td68qb96j7bqcj4cf41kc/T/ipykernel_35952/2228572355.py:1: InterpolationWarning: The test statistic is outside of the range of p-values available in the look-up table. The actual p-value is smaller than the p-value returned.

```
pvalue = tsa.stattools.kpss(resid.values())[1]
```

Necessary properties of residuals: uncorrelatedness

```
In [16]: corr_model = ARIMA(p=1, d=0, q=0, seasonal_order=(0, 0, 0, 0))
         fcast = corr_model.historical_forecasts(
             air_pax, forecast_horizon=1, start_format="position", show_warnings=F
         )
         resid = air_pax[fcast.start_time():] - fcast
```

```
In [18]: fig, ax = plt.subplots(1,3,figsize=(16,9))
         air_pax.plot(ax=ax[0]);
         fcast.plot(ax=ax[0]);
         ax[1].stem(resid.values());
         plot_acf(resid, max_lag=12*5, axis=ax[2])
```



```
In [19]: sts.ttest_1samp(resid.values(), 0).pvalue
```

```
Out[19]: array([0.04967464])
```

```
In [20]: tsa.stattools.kpss(resid.values())[1]
```

/var/folders/33/j0cl7y453td68qb96j7bqcj4cf41kc/T/ipykernel_35952/3591776226.py:1: InterpolationWarning: The test statistic is outside of the range of p-values available in the look-up table. The actual p-value is greater than the p-value returned.

```
tsa.stattools.kpss(resid.values())[1]
```

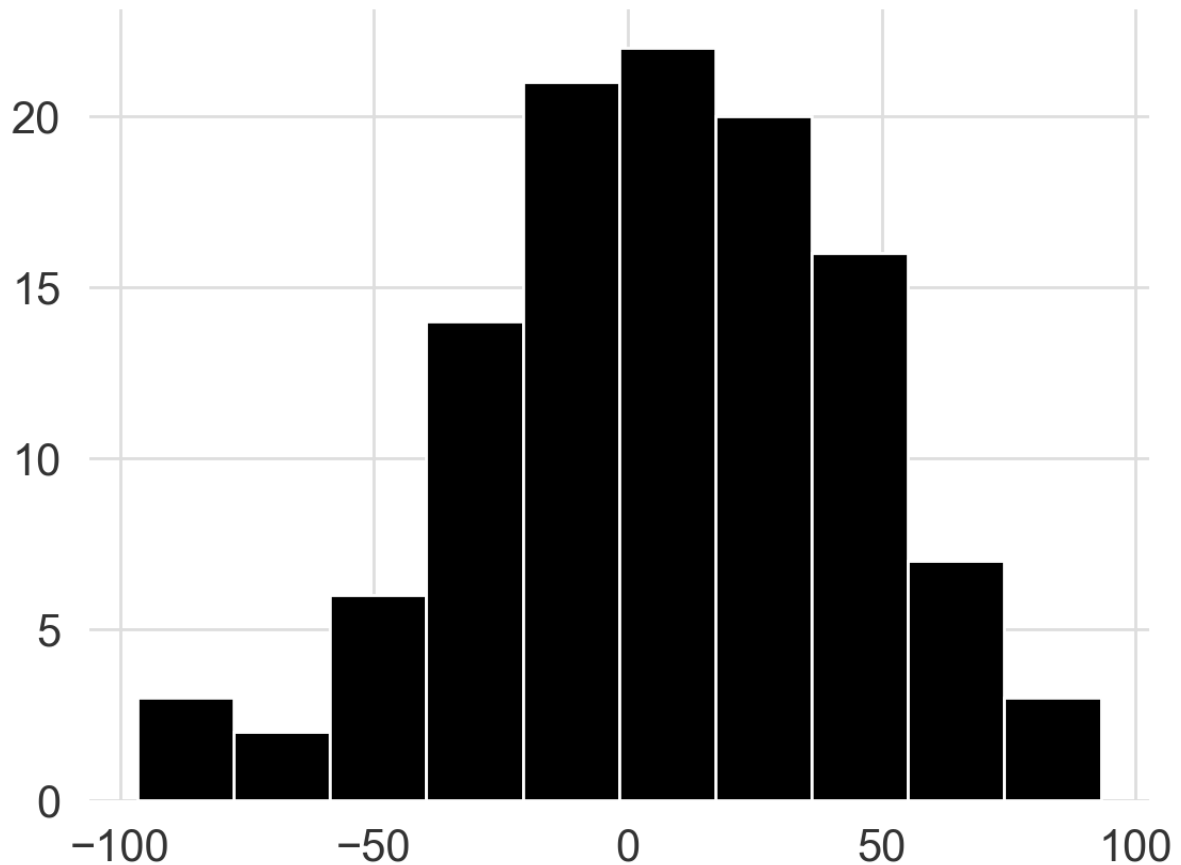
```
Out[20]: 0.1
```

```
In [21]: sm.stats.acorr_ljungbox(resid.values(), model_df=1)["lb_pvalue"].max()  
rejected = pvalue < 0.05  
print("there is", "no autocorrelation" if not rejected else "autocorrelation")
```

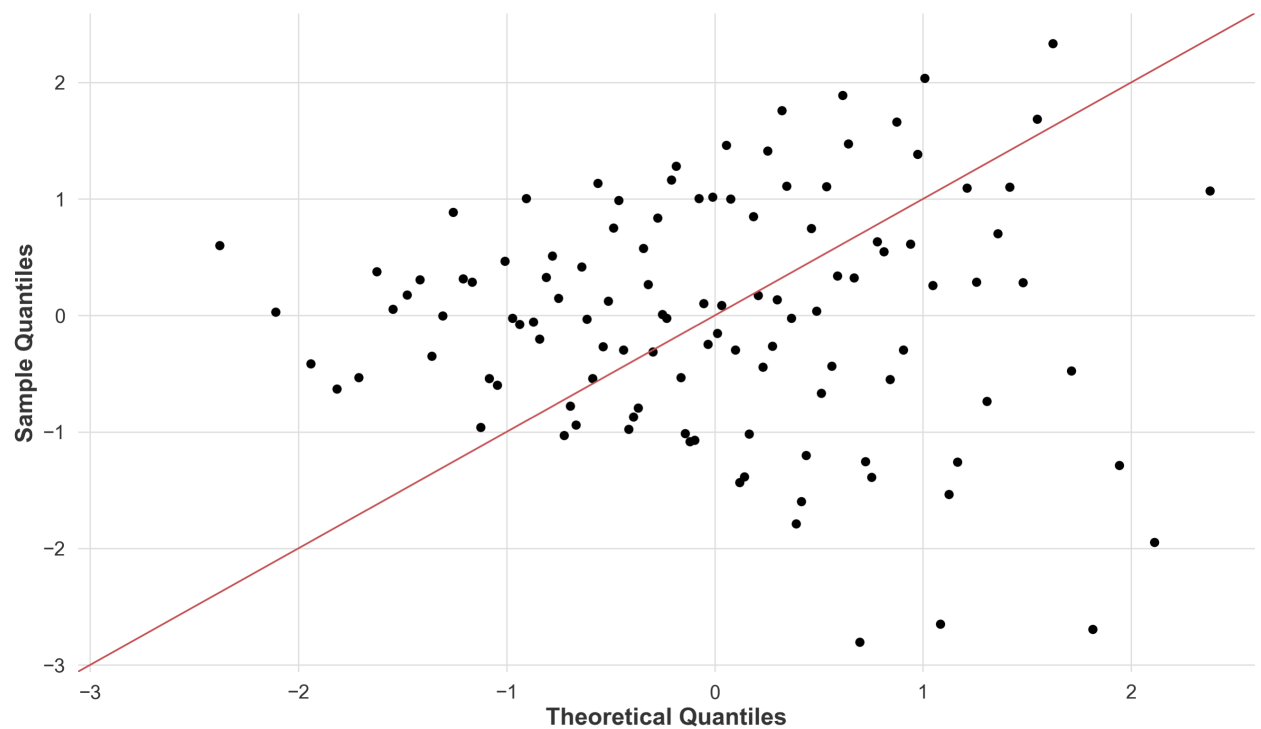
there is autocorrelation

Desired properties of residuals: normality

```
In [22]: plt.hist(resid.values());
```



```
In [23]: fig, ax = plt.subplots(figsize=(16,9))
sm.qqplot(resid.values(), ax=ax, fit=True, line="45");
```



```
In [30]: pvalue = sts.shapiro(resid.values()).pvalue
rejected = pvalue < 0.05
print("normal" if not rejected else "not normal")
```

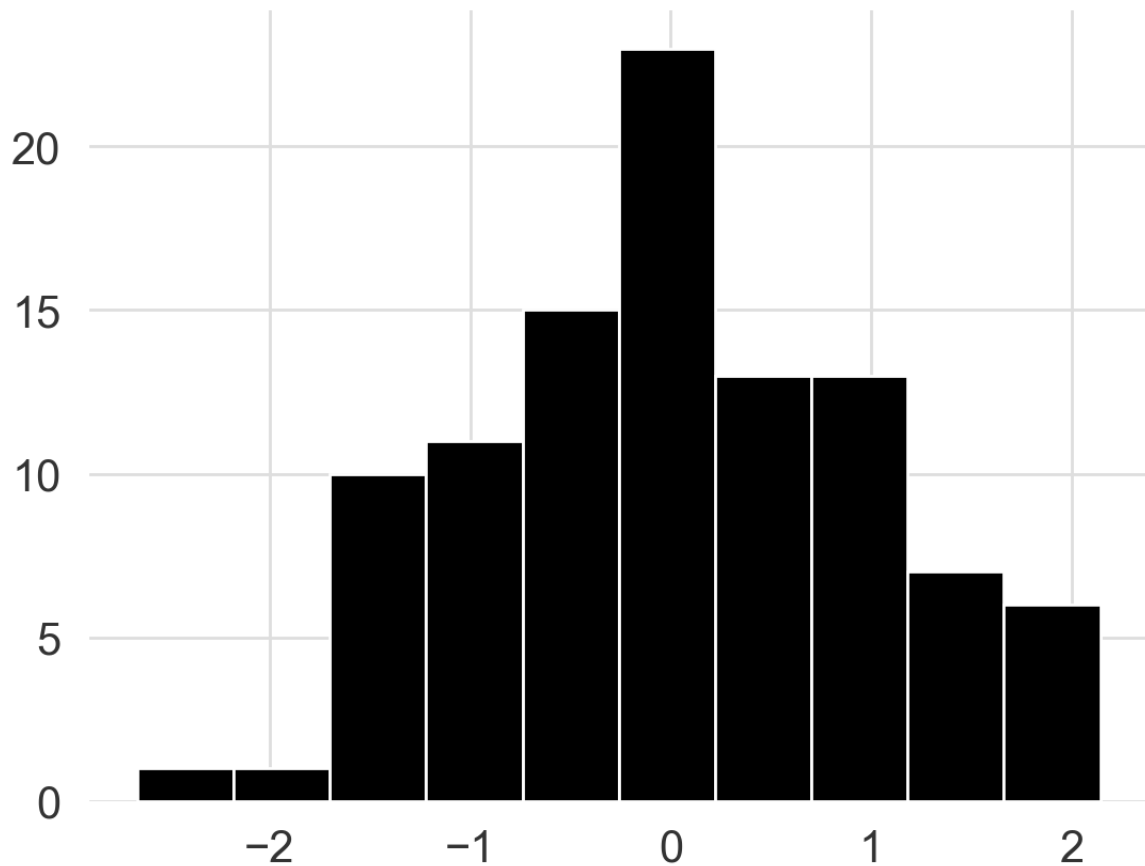
normal

```
In [31]: pvalue
```

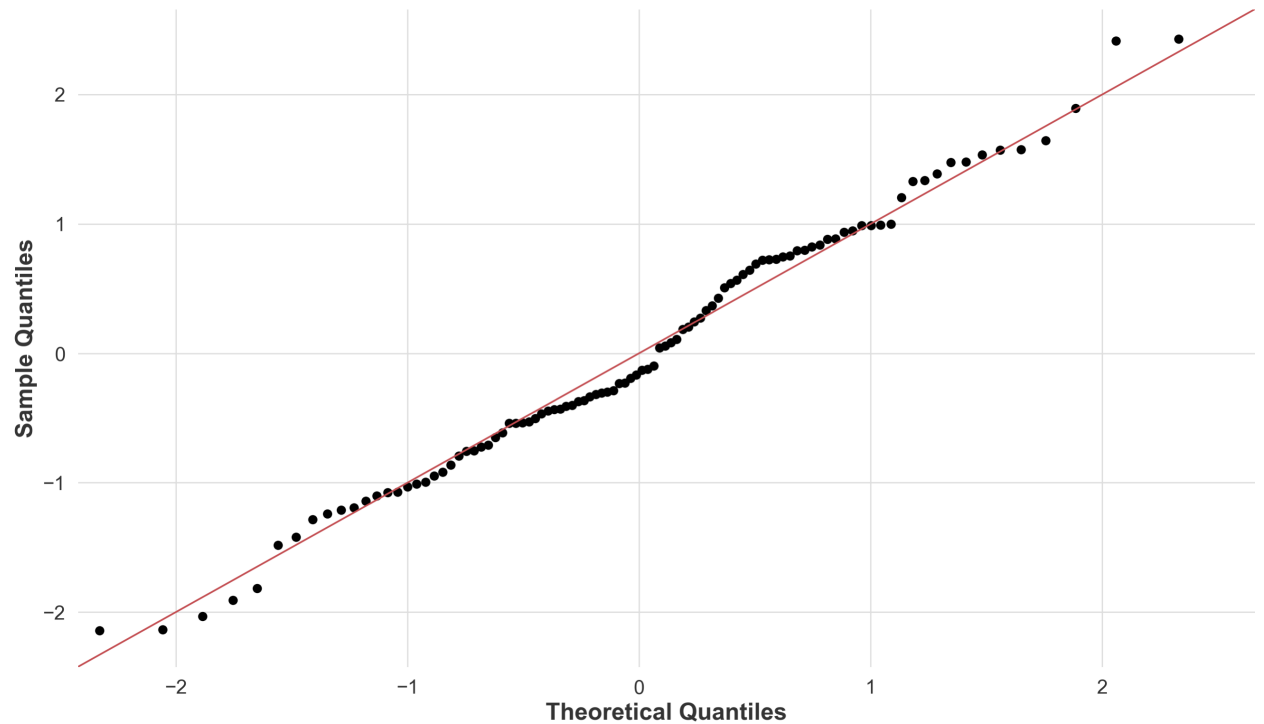
```
Out[31]: 0.5282973647117615
```

```
In [26]: dummy_resid = sts.norm.rvs(size=100)
```

```
In [28]: plt.hist(dummy_resid);
```



```
In [25]: fig, ax = plt.subplots(figsize=(16,9))
sm.qqplot(dummy_resid, ax=ax, fit=True, line="45");
```



```
In [32]: pvalue = sts.shapiro(dummy_resid).pvalue
         rejected = pvalue < 0.05
         print("normal" if not rejected else "not normal")
```

normal

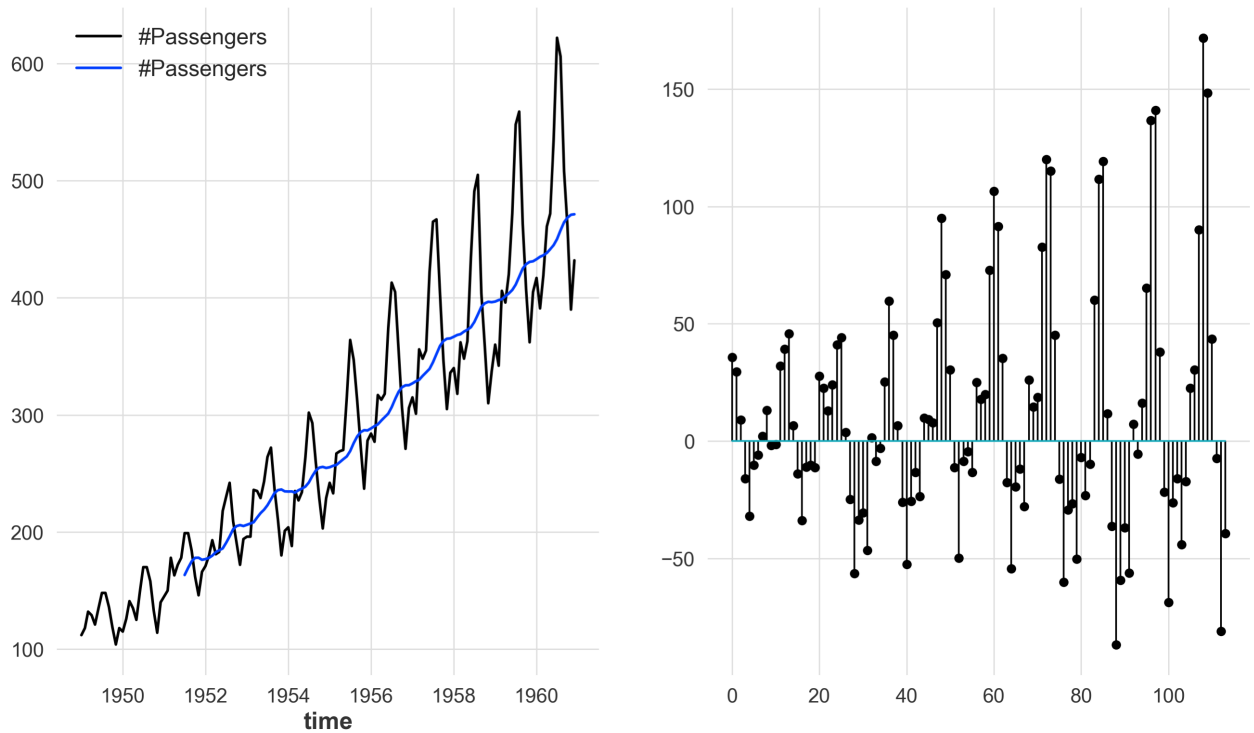
```
In [33]: pvalue
```

```
Out[33]: 0.8046566843986511
```

Desired properties of residuals: homoscedacity

```
In [34]: const_model = ARIMA(p=0, d=0, q=0, seasonal_order=(0, 0, 0, 0), trend="ct")
         fcast = const_model.historical_forecasts(
             air_pax, start=2, forecast_horizon=1, start_format="position", show_w
         )
         resid = air_pax[fcast.start_time():] - fcast
```

```
In [35]: fig, ax = plt.subplots(1,2,figsize=(16,9))
         air_pax.plot(ax=ax[0]);
         fcast.plot(ax=ax[0]);
         ax[1].stem(resid.values());
```



```
In [ ]: pvalue = sm.stats.diagnostic.het_breuschpagan(resid.values(), ...).pvalue
         rejected = pvalue < 0.05
         print("normal" if not rejected else "not normal")
```

AutoARIMA

```
In [46]: from darts.models import AutoARIMA
```

```
In [47]: from pmdarima import AutoARIMA as pmdAutoArima
```

```
In [ ]: pmdAutoArima()
```

```
In [51]: auto_arima = AutoARIMA(seasonal=True, m=12) # D = 1
         auto_arima.fit(air_pax);
```

```
In [54]: auto_arima.model.model_.order
```

```
Out[54]: (2, 1, 1)
```

```
In [55]: auto_arima.model.model_.seasonal_order
```

```
Out[55]: (0, 1, 0, 12)
```

```
In [56]: auto_arima.model.model_.trend
```

```
In [59]: good_model = ARIMA(p=2, d=1, q=1, seasonal_order=(0, 1, 0, 12), trend="n"
         fcst = good_model.historical_forecasts(
             air_pax, forecast_horizon=1, start_format="position", show_warnings=F
```

[illegible]

```
warn('Non-stationary starting autoregressive parameters'  
/Users/nstulov/miniconda3/envs/msai/lib/python3.12/site-packages/statsmode  
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[illegible]

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[illegible]

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[illegible]

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```

```
In [60]: fig, ax = plt.subplots(1,3,figsize=(16,9))
air_pax.plot(ax=ax[0]);
```

```
fcast.plot(ax=ax[0]);  
ax[1].stem(resid.values());  
plot_acf(resid, max_lag=12*5, axis=ax[2])
```

